

DBLOK7 Genes in Motion: Bioinformatics and Systems Biology – Summer School

July 13 - Jul 24

Welcome!

Schedule

Monday-Friday

9:00-12:00, with a short coffee break

Schedule (the first week)

July 13th - From RNA Molecules to Aligned Reads: Understanding Transcriptomic Data

July 14th - Gene Regulatory Networks: Data driven introduction

July 15th - Boolean Modelling: Introduction

July 16th - Boolean Modelling: Building models from requirements

July 17th - Boolean Modelling and GRNs: A case study

Schedule (the second week)

July 20th - Differential Expression: How We Compare Cell Types

July 21th - Differential Expression: Practical workflow - alignment and counting

July 22th - Differential Expression: bulk RNA-seq analysis with hisat2, featureCounts and pyDeSeq2

July 23th - Transcriptome annotation and assembly

July 24th - RNA modifications and final discussion

The teachers



Mariateresa Mazzetto



David Šafránek



Sam Pastva



Matej Lexa



Monika Čechová



Honza Kotrs

Course requirements

Learning outcomes are assessed through active participation in practical sessions, regular attendance, and group discussions. During the discussion sessions, students demonstrate and explain the computational analyses they performed, interpret their results, and discuss the underlying biological concepts and methodological choices. Successful course completion requires regular attendance and active participation in group discussions demonstrating achievement of the intended learning outcomes.

Lunch around the Faculty of Informatics

<https://obed.cucin.eu/gotex/>

Coffee breaks

3rd floor kitchen

(Please note using of the coffee machine for the accounting purposes)



Please ask questions!

Let's get started!

guest00

we shop and work

<https://github.com/sybila/bioinformatics-summer-school>